

In this Standard the comma has been adopted as the decimal sign.

INDEX

1.	<u>SCOPE</u>	2
2.	<u>APPLICATION</u>	2
3.	<u>APPLICABLE DOCUMENTS</u>	2
4.	<u>SURFACE CLASSIFICATION</u>	3
5.	<u>REQUIREMENTS</u>	7
5.1	SURFACE WAVINESS	7
5.2	SURFACE ROUGHNESS	9
5.3	BASIC CONTOUR LINES.....	9
5.4	STEPS AND GAPS.....	10
5.4.1	Fasteners	11
5.4.2	Skin Joints, Access Panels and Doors.....	13
5.4.3	Doors and Windows in Pressurized Areas.....	14
5.4.4	Control Surfaces, Flaps and Slats	15
5.5	CONTROL SURFACES, SLAT AND FLAP ANGULAR DEFLECTION TOLERANCES	17
5.6	COMPONENT ANGULAR TOLERANCES	18
5.7	SPECIAL AERODYNAMIC REQUIREMENTS	18
6.	<u>MAXIMUM ALLOWABLE DEFORMATIONS DUE TO IN-FLIGHT LOADS</u> .	19
7.	<u>QUALITY ASSURANCE PROVISIONS</u>	22
8.	<u>GENERAL NOTES</u>	22
8.1	DRAWING CALLOUT.....	22
9.	<u>BIBLIOGRAPHY</u>	22

NE 02-046

AERODYNAMIC SMOOTHNESS – REQUIREMENTS FOR ERJ 190 AND ERJ 195

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ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 1 of 22

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1. SCOPE

This Standard establishes the aerodynamic smoothness requirements for the ERJ 190 and ERJ 195 aircraft skin contours, such as steps, gaps, installation of fasteners, angular tolerances for control surfaces, slats and flaps, component angular tolerances and other configurations established on the production drawings.

NOTES

- 1) Those tolerances and conditions of applicability are described in Item 4 and Item 5.
- 2) The maximum allowable deformation of external surfaces submitted to in-flight normal operational loads is presented in Item 6.
- 3) In order to verify if the aerodynamic requirements are being correctly controlled during aircraft manufacture, the Aerodynamics Department is supposed to evaluate and approve the inspection methods and procedures that will be developed to control the relevant characteristics that are defined in this Standard.

In order to guarantee that the aerodynamic requirements are being correctly indicated in manufacturing and assembly drawings and appropriately monitored during aircraft manufacture process, the Embraer Aerodynamics Engineering Group is supposed to evaluate and approve the applicable inspection methods and test procedures that will be adopted to control all relevant characteristics that are defined in this document.

2. APPLICATION

This Standard is applicable to all external surfaces of the ERJ 190 and ERJ 195 aircraft, unless otherwise specified on production drawings.

3. APPLICABLE DOCUMENTS

Embraer Standards

- NE 03-004 - Surface Roughness.
- NE 03-006 - Rivets and Pins - Drilling, Countersinking and Dimpling.
- NE 03-068 - Bolts And Screws - Drilling, Countersinking and Dimpling for Installation.
- NE 03-085 - Surface Roughness – Inspection.
- NE 07-038 - EMB-145, EMB-314, EMB-390, EMB-500/505, EMB-550 and ERJ 170/190 Visual Defects - Rework and Acceptance Criteria for External Areas.

NE 02-046

AERODYNAMIC SMOOTHNESS – REQUIREMENTS FOR ERJ 190 AND ERJ 195

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 2 of 22

4. SURFACE CLASSIFICATION

The aircraft external surfaces are divided into three areas ("IA", "IB" and "II"), according to aerodynamic smoothness, to facilitate the manufacturing and inspection procedures. The definitions of the extent of each area are given as follows in Table 1. See Figure 1, Figure 2 and Figure 3.

TABLE 1 : CRITICAL AND NON-CRITICAL AREAS

AREAS			DESCRIPTION
CRITICAL	“IA”	Fuselage	– Radome.
		Nacelle	– From lip of the inlet cowl to 406,4 mm (16 in) aft (internal and external surfaces.
		Pylon Fairing	– The entire pylon fairing external surface.
		Wing	– From the wing-fuselage intersection up to station Y = 5120 mm, from the leading edge up to the spar 3, upper and lower surfaces.
			– From station Y = 5120 mm to the wing tip, from the leading edge up to spar 2, upper and lower surfaces.
		Winglet	– From winglet root to tip.
			– From the leading edge up to aft spar, upper and lower surfaces.
		Slats	– The entire outer surface of the retracted slats
		Flaps	– From the leading edge up to 40% of the local chord for all flap panels.
	Horizontal Stabilizer	– From the leading edge up to front spar, upper and lower surfaces.	
	Vertical Tail	– From the leading edge up to front spar.	
	“IB”	Fuselage	– Forward fuselage, except radome.
		Horizontal Stabilizer	– From front spar to trailing edge, upper and lower surfaces.
		Vertical Tail	– From front spar to trailing edge.
		Elevator	– The entire elevator external surface.
		Rudder	– The entire rudder external surface.
		WFF	– The entire wing-to-fuselage fairing (WFF) external surface.
“II” (1)	-	– The remaining external surfaces.	

(1) The area "II" is called "non-critical".

NE 02-046

AERODYNAMIC SMOOTHNESS – REQUIREMENTS FOR ERJ 190 AND ERJ 195

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 3 of 22

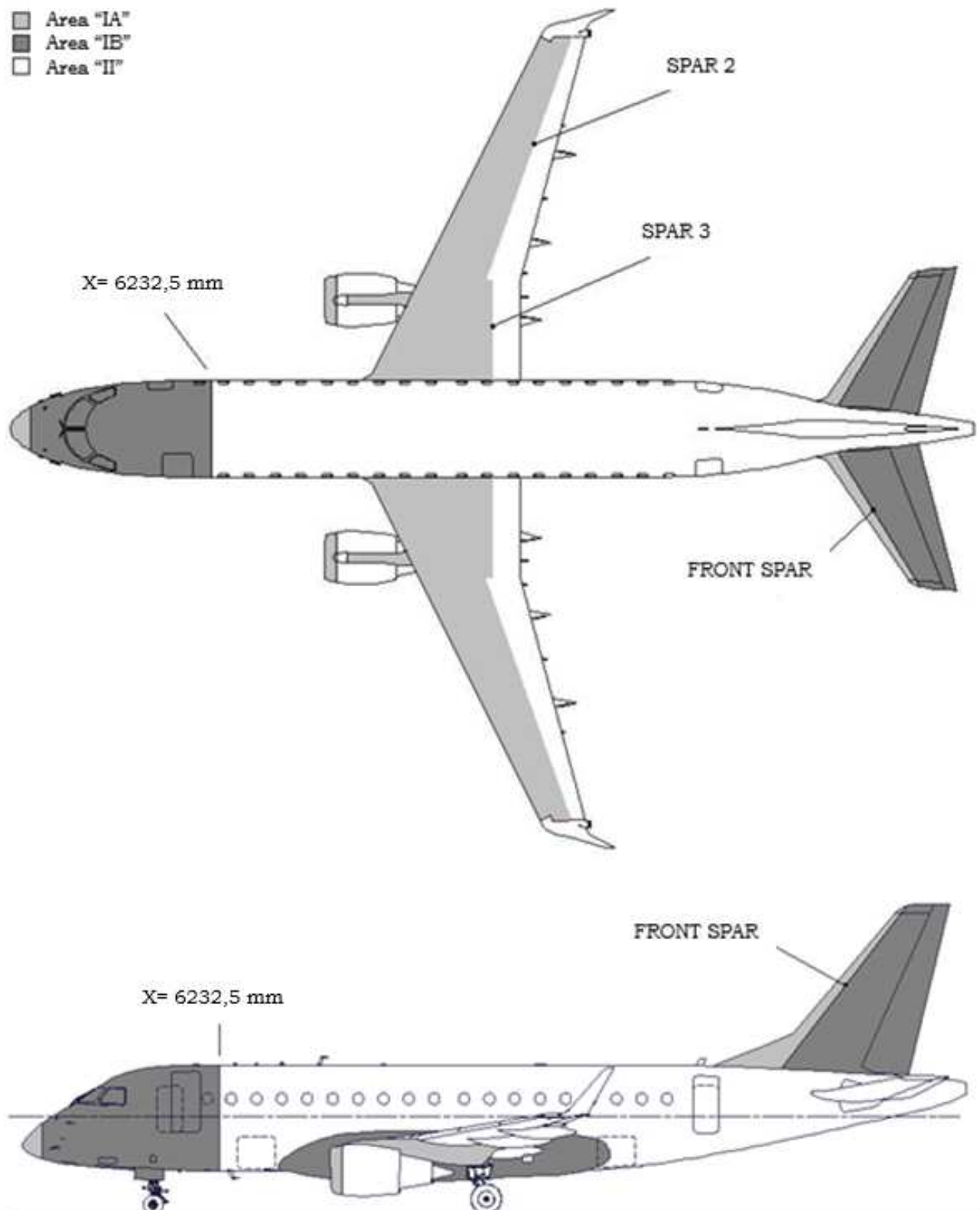


FIGURE 1 : AERODYNAMIC AREA CLASSIFICATION – ILLUSTRATION

NE 02-046

**AERODYNAMIC SMOOTHNESS –
REQUIREMENTS FOR ERJ 190 AND ERJ 195**

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 4 of 22

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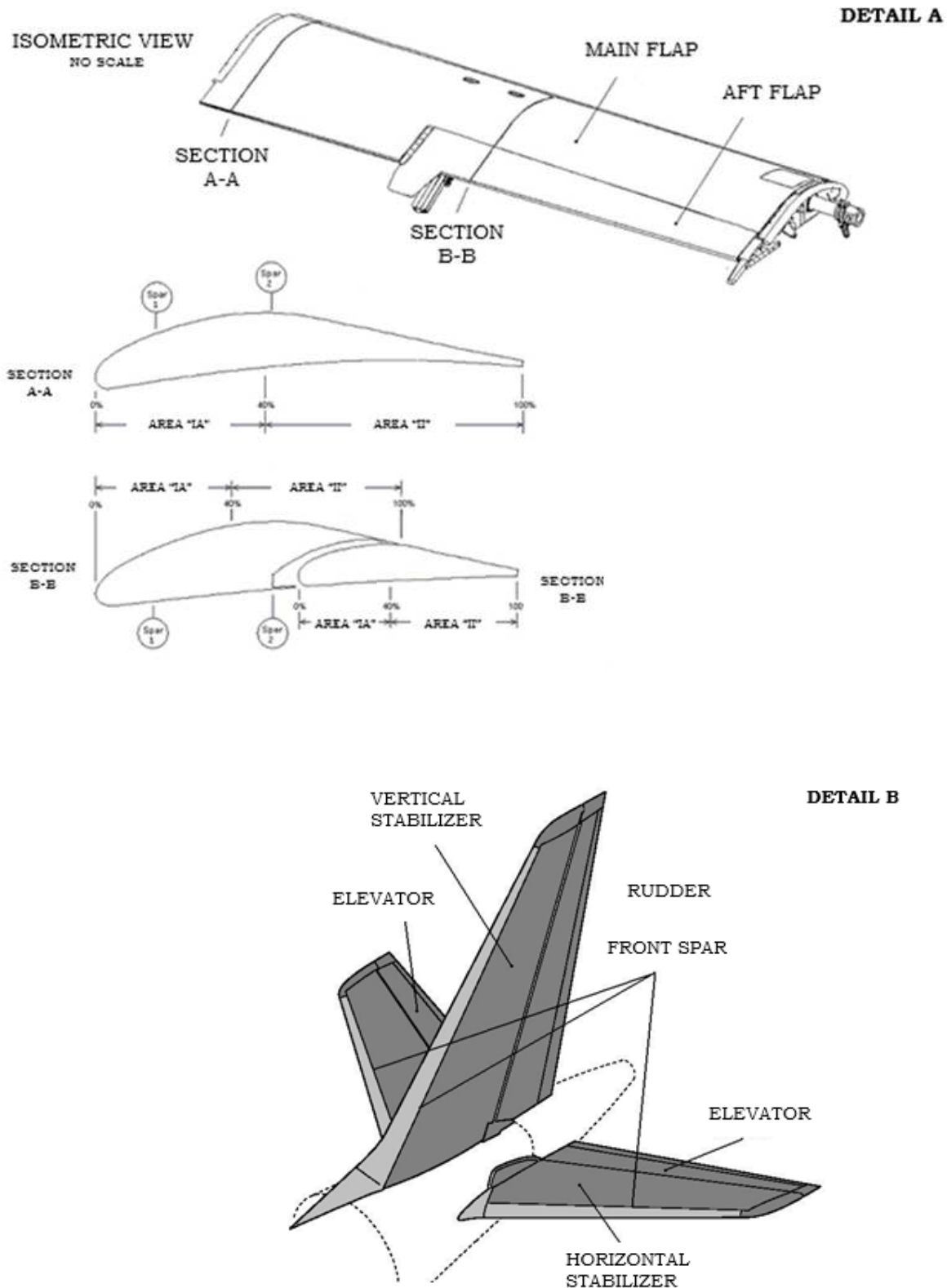


FIGURE 2 : AERODYNAMIC AREA CLASSIFICATION – DETAILS A AND B

NE 02-046

**AERODYNAMIC SMOOTHNESS –
REQUIREMENTS FOR ERJ 190 AND ERJ 195**

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 5 of 22

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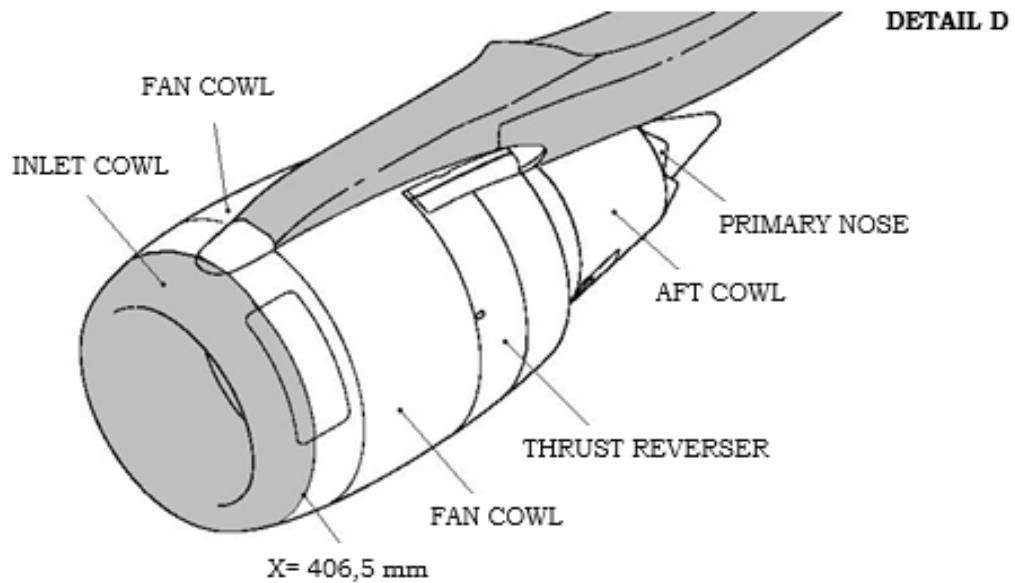
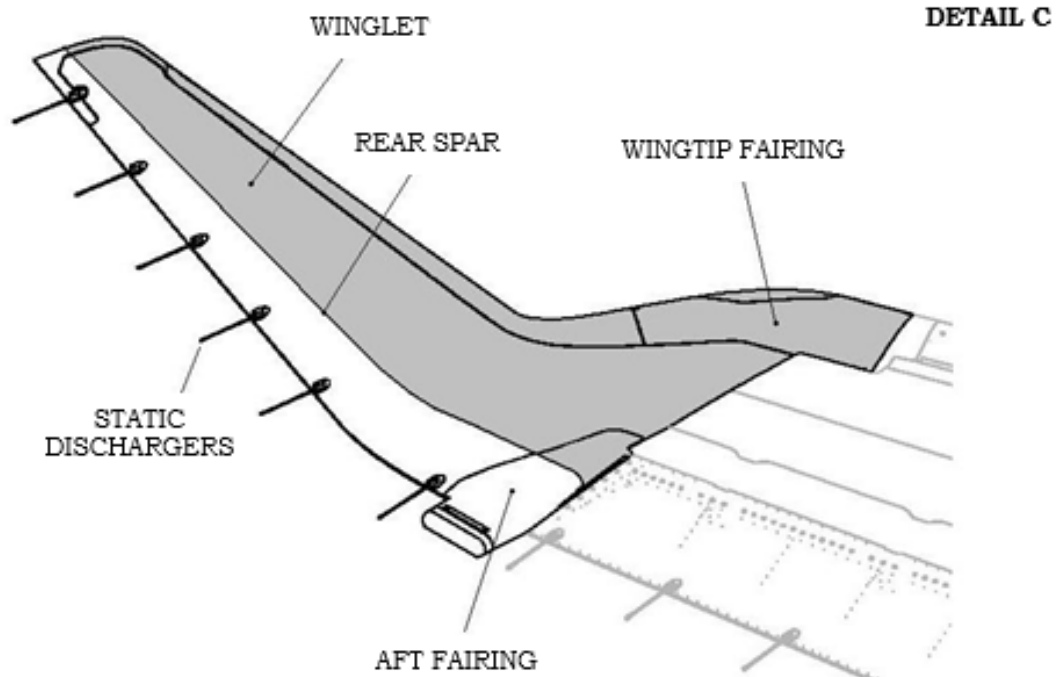


FIGURE 3 : AERODYNAMIC AREA CLASSIFICATION – DETAILS C AND D

NE 02-046

**AERODYNAMIC SMOOTHNESS –
REQUIREMENTS FOR ERJ 190 AND ERJ 195**

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 6 of 22

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5. REQUIREMENTS

The requirements listed hereinafter are applicable to the areas "IA", "IB" and "II", unless otherwise specified. See Item 3.

5.1 SURFACE WAVINESS

Single, compound (total or partial) and periodic waves shown in Figure 4, Figure 5 and Figure 6 shall meet the aerodynamic smoothness requirements specified in Figure 7. See Item 5.3.

NOTE

- 1) The wavelength shall be measured parallel to flight direction. Waves on riveting areas (normal to flight direction) only affect visual aspect. See NE 07-038.
- 2) For waves located at the edge of a part, where the measurement using two-point support for a straight-edge parallel to flight direction is not possible, the wavelength may be determined from consecutive spanwise measurements, starting from the edge, performed in different positions moving the measuring device chordwise until the wave height returns to zero in order to reconstruct the longitudinal wave. In such cases, the wave height (H) shall be the largest wave height measured, and the wavelength (L) shall be the total spanwise measured length multiplied by two.

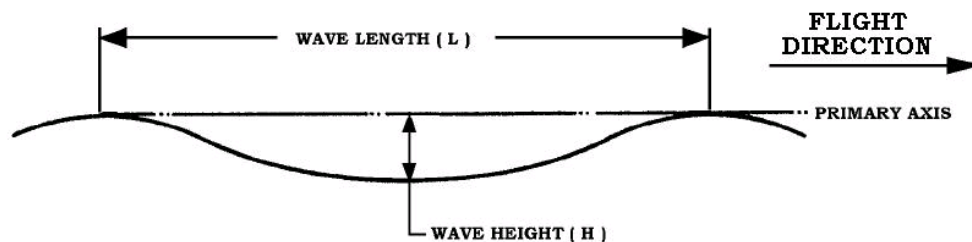


FIGURE 4 : WAVE TYPES – SINGLE WAVE

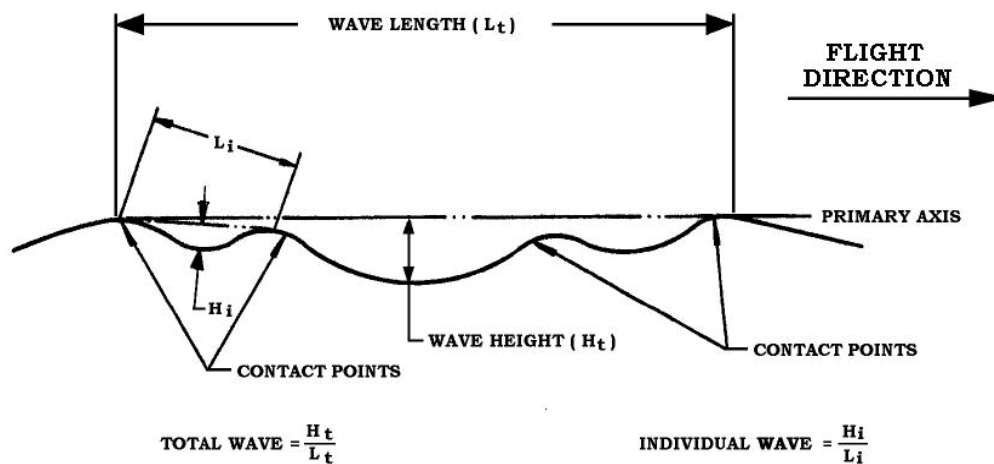


FIGURE 5 : WAVE TYPES – COMPOUND WAVE

NE 02-046

**AERODYNAMIC SMOOTHNESS –
REQUIREMENTS FOR ERJ 190 AND ERJ 195**

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 7 of 22

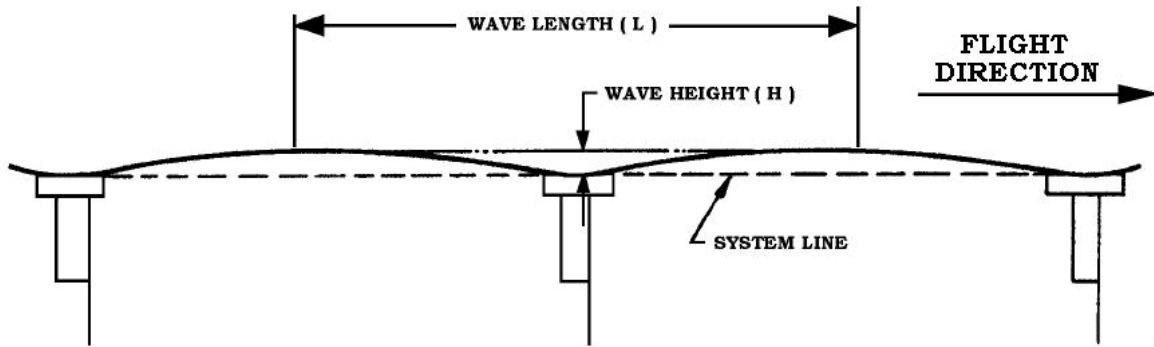
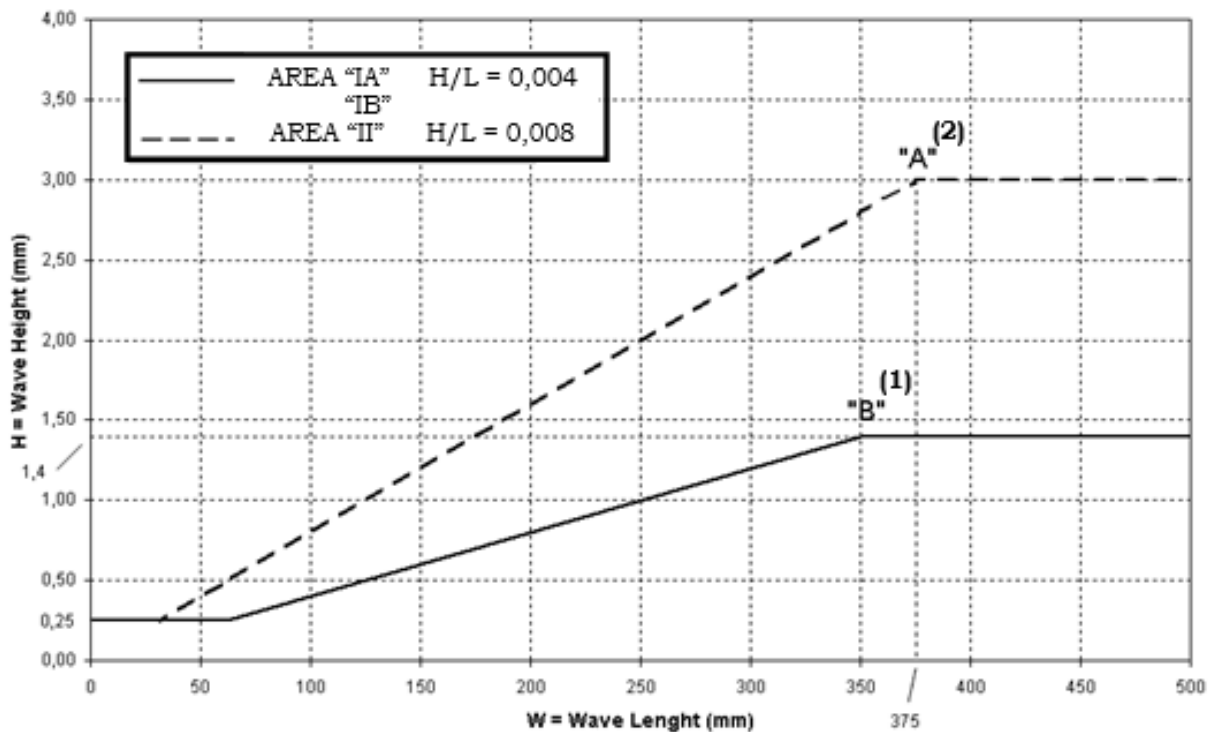


FIGURE 6 : WAVE TYPES – PERIODIC WAVE



- (1) Maximum wave height (H) value to area “IA” and “IB” is 1,4 mm for wavelength (W) greater than 350 mm.
 (2) Maximum wave height (H) value to area “II” is 3,0 mm for wavelength (W) greater than 375 mm.

Point “A” : $L_A = \frac{H_A}{0,008} = \frac{3}{0,008} = 375mm$

Point “B” : $H_B = L_B \times 0,004 = 350 \times 0,004 = 1,4mm$

FIGURE 7 : MAXIMUM ALLOWABLE HEIGHTS FOR WAVES

NE 02-046

**AERODYNAMIC SMOOTHNESS –
 REQUIREMENTS FOR ERJ 190 AND ERJ 195**

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 8 of 22

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5.2 SURFACE ROUGHNESS

The roughness of the aerodynamic surfaces is related to the depth of grooves and size of grains distributed along each surface, depending on the finishing process.

The surface finish roughness for external surfaces shall not exceed 3,2 μm (125 μin) Ra with a 2,50 mm (0,10 in) of sampling length (cutoff). For Ra and sampling length definitions. See NE 03-004 (or NE 03-085 for new drawings).

5.3 BASIC CONTOUR LINES

The aircraft contour lines deviations shall be within the tolerances indicated in Table 2 and Figure 8. The requirements of both Item 5.1 and Item 5.3 shall be met.

TABLE 2 : BASIC CONTOUR LINE TOLERANCE

AREA	TOLERANCE (mm)
"IA"	1,4
"IB"	2,2
"II"	3,0

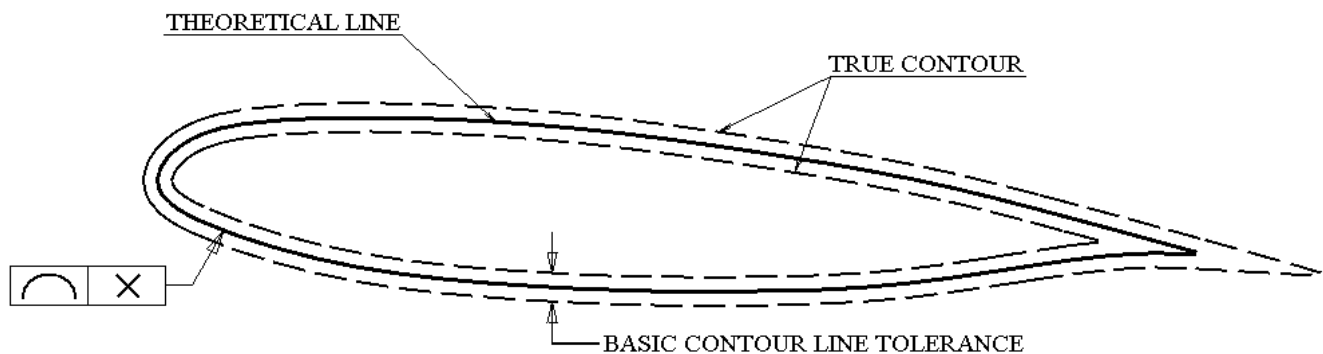


FIGURE 8 : BASIC CONTOUR LINE TOLERANCE

The contour line tolerance requirement is specified for the entire span of aircraft components; however, specific control sections shall be defined in agreement with the Embraer Aerodynamics Engineering Group.

NE 02-046

**AERODYNAMIC SMOOTHNESS –
REQUIREMENTS FOR ERJ 190 AND ERJ 195**

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 9 of 22

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5.4 STEPS AND GAPS

Multiple steps and gaps are present on the external surface of an aircraft. The influence of these discrepancies on aerodynamic performance may be markedly strong, thus the need for controlling them.

The definition of normal and parallel steps and gaps is shown in Figure 9. Gaps and steps between the movable surfaces (ailerons, elevators, rudder, spoilers, etc.), doors and adjacent areas are discussed separately as special cases.

NOTE

The sign convention for normal steps is presented in Figure 10 are:

"+" means steps "against" the flight direction; and

"-" means "favorable" steps.

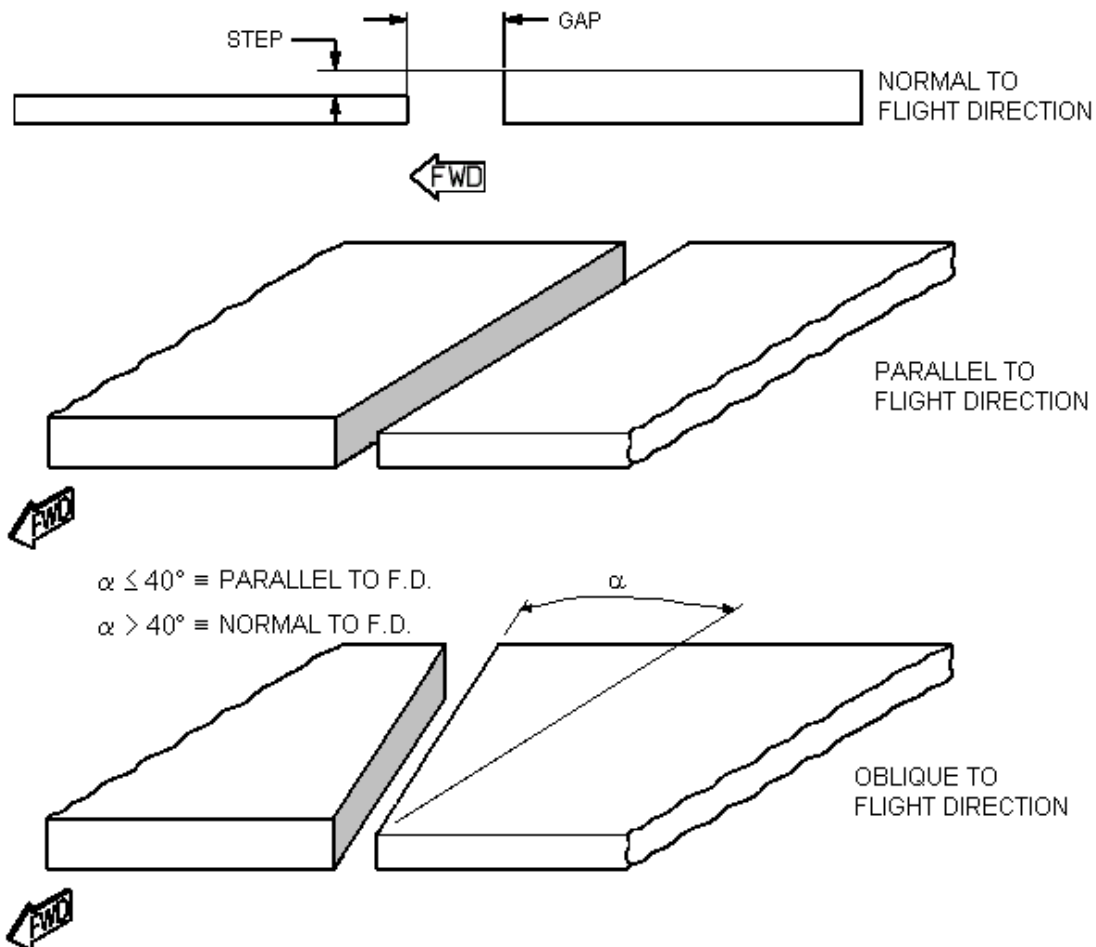


FIGURE 9 : STEPS AND GAPS DEFINITIONS

NE 02-046

**AERODYNAMIC SMOOTHNESS –
REQUIREMENTS FOR ERJ 190 AND ERJ 195**

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 10 of 22

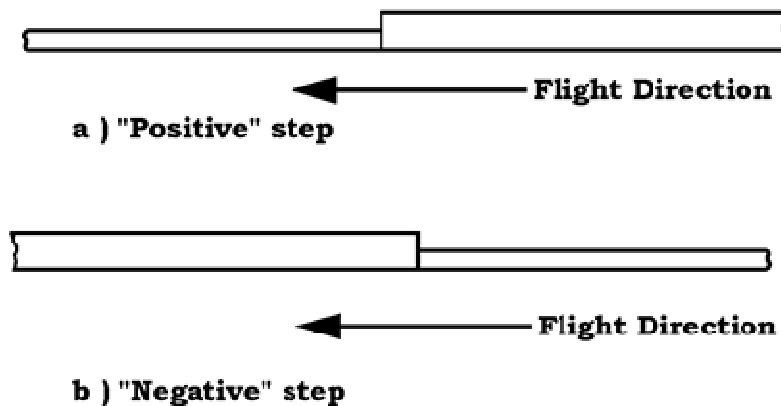


FIGURE 10 : SIGN CONVENTION FOR NORMAL STEPS

5.4.1 Fasteners

- a) protuberant head fasteners shall not be used on area "IA" and "IB". The use of protuberant head fasteners on area "II" shall be subject of analysis and approval of Embraer Aerodynamics Group on a case-by-case basis;
- b) countersunk head fasteners shall be according to Figure 11.

NOTES

- 1) Fasteners not complying with aerodynamic smoothness tolerances herein specified may be accepted provided that they do not exceed 10% of the fasteners that are installed in the airplane and that they are not located in areas "IA" and "IB".
- 2) The maximum values for upper and lower deviations shall be according to NE 03-006 for rivets and pins or NE 03-068 for bolts and screws.

NE 02-046

AERODYNAMIC SMOOTHNESS – REQUIREMENTS FOR ERJ 190 AND ERJ 195

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 11 of 22

COUNTERSUNK FASTENER

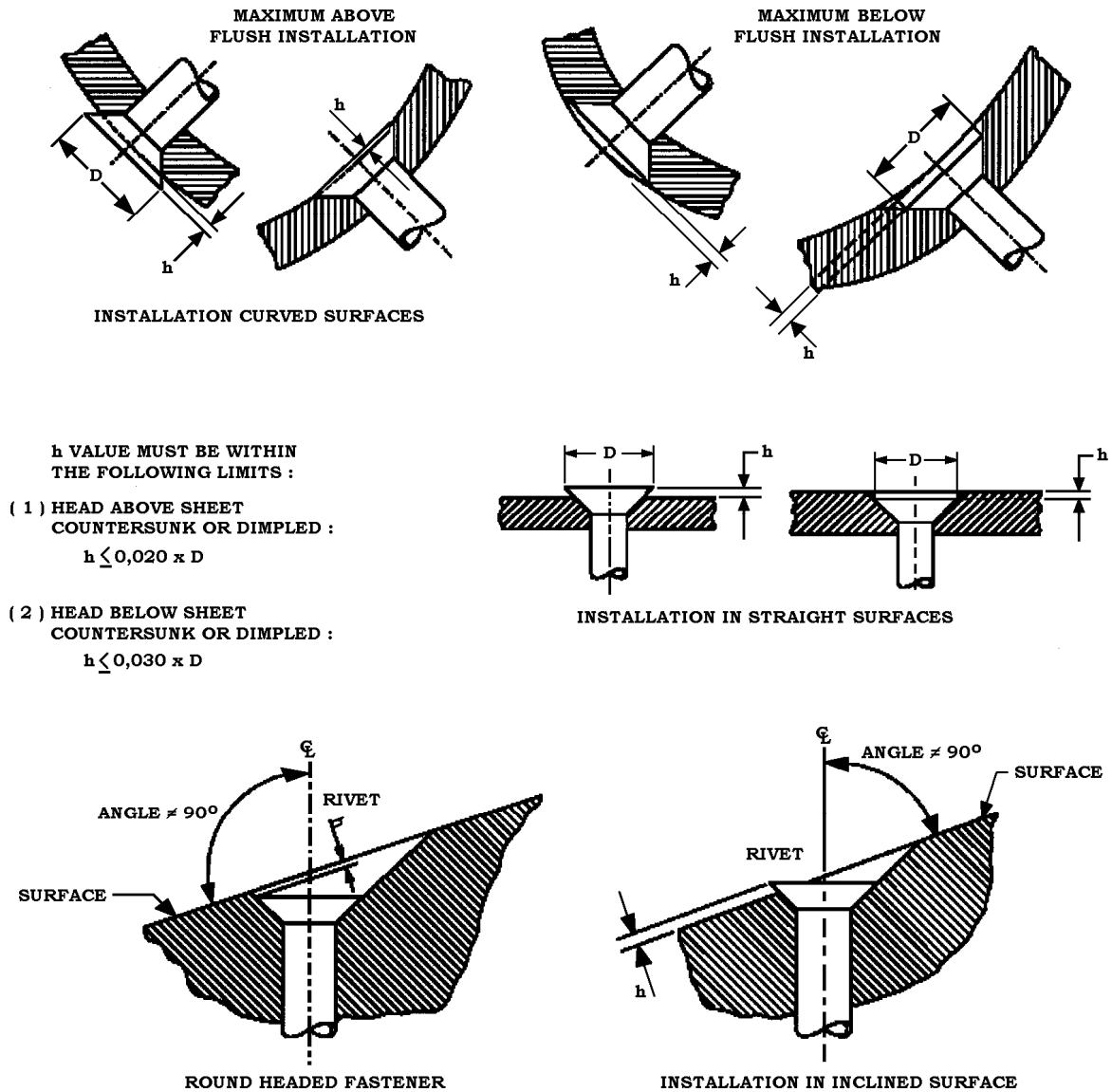


FIGURE 11 : ALLOWABLE TOLERANCES FOR COUNTERSUNK HEAD FASTENER POSITION IN REGARD TO SHEET SURFACE

NE 02-046

**AERODYNAMIC SMOOTHNESS -
 REQUIREMENTS FOR ERJ 190 AND ERJ 195**

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 12 of 22

This document shall not be used, published nor communicated to third parties or reproduced without authorization of the issuing company.

5.4.2 Skin Joints, Access Panels and Doors

The maximum allowable steps and gaps for joints (for example: plate splices, access doors, inspection doors, etc.) shall be according to Table 3. Allowable steps and gaps for doors in pressurized areas are given in Item 5.4.3. Definition of steps and gaps is given in Figure 12.

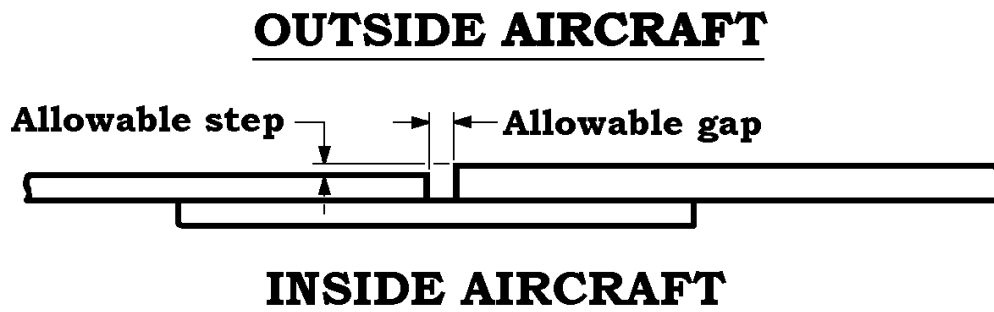


FIGURE 12 : ALLOWABLE TYPE OF SKIN JOINTS BETWEEN FIXED SURFACES

TABLE 3 : MAXIMUM ALLOWABLE STEPS AND GAPS FOR SKIN JOINTS

JOINT TYPE	AREA	STEP (mm)		GAP (mm) (3)	
		NORMAL TO F.D. (1)	PARALLEL TO F.D. (1)	NORMAL TO F.D. (1)	PARALLEL TO F.D. (1)
FIXED SURFACES (5) (7)	"IA"	± 0,5	0,5	1,6	1,6
	"IB"	± 0,5	0,5	1,6	1,6
	"II"	± 0,8	0,8	2,0	2,0
SKIN AND BOLTED ACCESS DOORS AND INSPECTION PLATES	"IA"	± 0,5	0,5	2,0	2,0
	"IB"	± 0,5	0,5	2,0	2,0
	"II"	± 0,8	0,8	2,0	2,0
SKIN AND ALL OTHER KINDS OF ACCESS DOORS OR PLATES	"IA"	± 1,0	1,0	2,0	2,0
	"IB"	± 1,0	1,0	2,0	2,0
	"II"	± 1,0	1,0	3,0	3,0
NOSE LANDING GEARS DOORS (4)	"IB"	± 1,5	1,5	(8)	(8)
MAIN LANDING GEARS DOORS (4)	"IA"	± 1,5	1,5	(8)	(8)
WING-FUSELAGE FAIRING ACCESS PANELS	"IA"	± 0,5	0,5	6,5	6,5
RADOME AND FUSELAGE NOSE (6)	"IB"	± 1,1	N.A. (2)	2,0	N.A. (2)

(1) F.D. = Flight Direction.

(2) N.A. = Not Applicable.

(3) All gaps between fixed surfaces shall be filled with sealant and smoothed. The maximum allowable ridge or depression between sealant and skin surface shall be according to Figure 13.

The notes of Table 3 continue.

NE 02-046

AERODYNAMIC SMOOTHNESS –
REQUIREMENTS FOR ERJ 190 AND ERJ 195

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 13 of 22

The notes of Table 3 continued.

- (4) The landing gear door gaps shall be sealed in the retracted position.
- (5) The gap between the skin and the removable leading edge of the empennages shall be sealed and smoothed after each removal and installation.
- (6) Radome steps of up to 1,4 mm are allowed up to 20% of its perimeter.
- (7) Due to drag considerations, the use of lap joints shall be avoided, and it only may be implemented after Aerodynamics Department approval. The allowable steps for the lap joint areas will be the thickness of the skin plus the tolerance for fixed joints in each surface area ("IA", "IB" and "II").
- (8) See Main and Nose Landing Gear assembly drawings for gap tolerances.

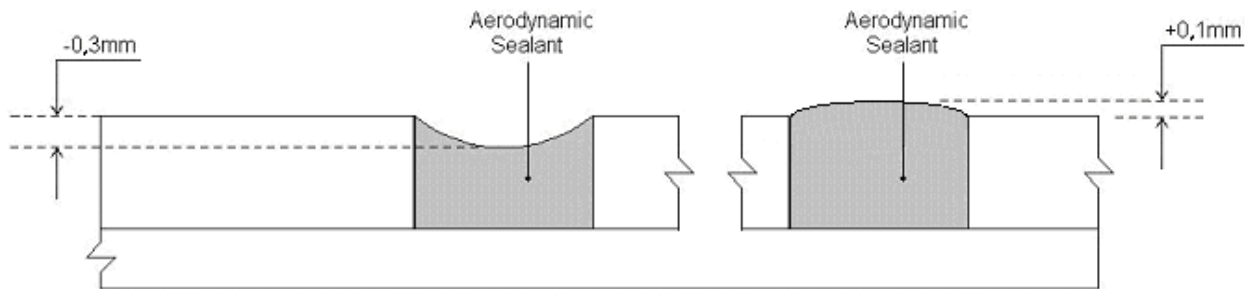


FIGURE 13 : MAXIMUM ALLOWABLE SEALANT RIDGE OR DEPRESSION

5.4.3 Doors and Windows in Pressurized Areas

The maximum allowable steps and gaps for doors and windows located in pressurized areas shall be according to Table 4.

TABLE 4 : MAXIMUM ALLOWABLE STEPS AND GAPS FOR DOORS AND WINDOWS IN PRESSURIZED AREAS (2)

ITEM	ALLOWABLE STEP (mm)		ALLOWABLE GAP (mm)
	INSIDE	OUTSIDE	
MAIN AND SERVICE DOORS (5)	1,5 (3)	1,5 (3)	6,0
BAGGAGE DOORS (5)	2,0 (4)	2,0 (4)	6,0
OVER WING EMERGENCY DOORS	1,5	1,5	4,0
WINDSHIELD (1)	0,5 (6)	0,5 (6)	4,5 (6)
DIRECT VISION WINDOWS	1,5 (7)	1,5 (7)	5,0 (7)
PASSENGER WINDOWS (1)	1,0	1,0	2,0

- (1) Gaps shall be filled in with sealant.
- (2) The above tolerances are for the aircraft pressurized.
- (3) Steps of up to $\pm 2,0$ mm are allowed up to 25% of main door perimeter and 20% of service door perimeter.
- (4) Baggage doors steps of up to $\pm 2,5$ mm are allowed up to 20% of its perimeter.

The notes of Table 4 continue.

NE 02-046		AERODYNAMIC SMOOTHNESS – REQUIREMENTS FOR ERJ 190 AND ERJ 195	
REV: K	ISSUE DATE: 30 Jun 26	REVISION REPORT: RR_NE 02-046K	PAGE: 14 of 22

The notes of Table 4 continued.

(5) The above specified step tolerances are for the main, service and baggage doors are not applicable at the bottom of the door if the scuff plate is installed. In this case, step control in this area is not required.

(6) Steps of up to $\pm 0,8$ mm and gaps up to 5,5 mm are allowed up to 20% of its perimeter.

(7) Steps of up to $\pm 2,0$ mm and gaps up to 5,5 mm are allowed up to 20% of its perimeter.

5.4.4 Control Surfaces, Flaps and Slats

The allowable steps and gaps and allowable deviations shall be as follows:

a) due to deviations in the manufacturing and assembly processes of the aircraft parts, gaps and steps not predicted during design may be found and shall be controlled, in order to minimize the aerodynamic performance degradation.

Table 5 presents the tolerance for steps and gaps between retracted control surfaces and fixed parts;

TABLE 5 : ALLOWABLE STEPS AND PRODUCTION GAPS BETWEEN CONTROL SURFACES, RETRACTED FLAPS AND SLATS AND FIXED PARTS

SURFACE	STEP (mm)		GAP (mm)	
	NORMAL TO F.D. (1)	PARALLEL TO F.D. (1)	NORMAL TO F.D. (1)	PARALLEL TO F.D. (1)
RETRACTED SLAT TO WING	$\pm 1,5$	2,0	2,0 (6)	(8)
RETRACTED SLAT TO SLAT	$\pm 1,5$	2,0	N.A. (2)	(8)
AILERON TO WING	$\pm 1,5$	1,5	N.A. (2)	(7)
AILERON SHROUD TO WING	$\pm 0,5$ (9)	$\pm 0,8$ (10)	(8)	(8)
SPOILERS TO WING	$\pm 1,5$	1,5	(8)	(8)
FLAP SHROUD TO WING	$\pm 0,5$ (9)	$\pm 0,8$ (10)	(8)	(8)
RETRACTED FLAP TO LOWER SHROUD	$\pm 1,5$	1,5	2,0	5,0
RETRACTED FLAP TO SPOILERS	$\pm 1,5$	1,5	N.A. (2)	(8)
RETRACTED MAIN FLAP TO AFT FLAP	$\pm 1,5$	1,5	N.A. (2)	7,0
RETRACTED FLAP TO TORQUE BOX	N.A. (2)	1,5	N.A. (2)	(8)
RETRACTED FLAP TO FLAP TRACK FAIRING	$\pm 1,5$	N.A. (2)	N.A. (2)	N.A. (2)
ELEVATOR TO HORIZONTAL STABILIZER	$\pm 1,5$ (11)	1,5	N.A. (2)	5,0
RUDDER TO VERTICAL STABILIZER	$\pm 1,5$	1,5	N.A. (2)	5,0

(1) F.D. = Flight Direction.

(2) N.A. = Not Applicable.

(3) Longitudinal and lateral gaps between the movable surfaces and other parts shall be aerodynamically sealed when the surface is in the nominal position.

(4) Longitudinal gap between the inboard and outboard flaps shall be sealed for all flap deflections.

The notes of Table 5 continue.

NE 02-046		AERODYNAMIC SMOOTHNESS – REQUIREMENTS FOR ERJ 190 AND ERJ 195	
REV: K	ISSUE DATE: 30 Jun 26	REVISION REPORT: RR_NE 02-046K	PAGE: 15 of 22

The notes of Table 5 continued.

- (5) Longitudinal gap between slat panels shall be sealed for all slat deflections.
- (6) Lateral gap between slat lower surface and wing shall be aerodynamically sealed when the surface is retracted.
- (7) The step between the control surface or flap and the aircraft fixed parts shall be measured according to Figure 14. The step is considered zero when the extended contour line of the fixed part matches exactly that of the control surface. The sign convention for the steps is defined in Figure 10.
- (8) See assembly drawings for gap requirements.
- (9) Steps up to $\pm 1,0$ mm are allowed in 20% of the interface area between shroud and wing skin.
- (10) Steps up to $\pm 1,0$ mm (parallel to flight direction) are allowed between the shroud and torque box.
- (11) Steps up to $\pm 2,0$ mm are allowed in 20% of the interface area between stabilizer and elevator.

b) some surfaces (such as deflected flaps and slats) require a large but well-defined gaps between them and the aircraft fixed parts. The purpose of these gaps is related to the aerodynamic performance of the surface. Table 6 presents the tolerance for gaps between deflected control surfaces and fixed parts.

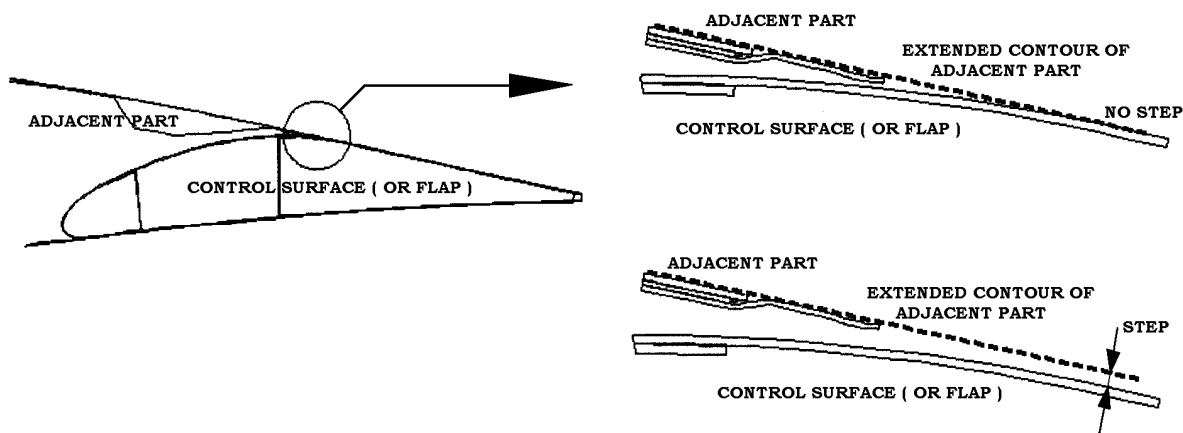


FIGURE 14 : STEPS BETWEEN CONTROL SURFACES AND FIXED PARTS

TABLE 6 : ALLOWABLE DEVIATIONS IN THE AERODYNAMIC GAPS BETWEEN DEFLECTED FLAPS, SLATS AND FIXED PARTS

SURFACE	GAP (mm) (1)	OVERLAP (mm) (1)
TAKEOFF SLATS TO WING	$\pm 1,0$ (2)	$\pm 2,0$
LANDING SLATS TO WING	$\pm 2,0$	$\pm 2,0$
INBOARD FLAP MAIN ELEMENT TO WING	$\pm 5,0$ (3)	$\pm 5,0$ (3)
INBOARD FLAP MAIN TO AFT ELEMENTS	$\pm 3,0$	$\pm 3,0$
OUTBOARD FLAP ELEMENT TO WING	$\pm 5,0$ (3)	$\pm 5,0$ (3)

The notes of Table 6 continue.

NE 02-046		AERODYNAMIC SMOOTHNESS – REQUIREMENTS FOR ERJ 190 AND ERJ 195	
REV: K	ISSUE DATE: 30 Jun 26	REVISION REPORT: RR_NE 02-046K	PAGE: 16 of 22

The notes of Table 6 continued.

- (1) The specified tolerances shall be observed for the entire range of surface deflection.
- (2) The gap tolerance for the slat #4 at the takeoff position is $\pm 1,5$ mm.
- (3) Gap and overlap values up to $\pm 6,0$ mm are allowed in one check position at the inboard flap and two check positions at the outboard flap if the difference between the out of tolerance value with its symmetrical position is lesser than $\pm 8,0$ mm.

5.5 CONTROL SURFACES, SLAT AND FLAP ANGULAR DEFLECTION TOLERANCES

The control surface deflection angle deviations may be divided into two types:

- Absolute Deviation: Angular deviation between the actual (measured) deflection and the nominal one.
- Symmetry Deviation: Difference between the actual (measured) deflections of the port and starboard parts of the same control surface, slat and flap, unless otherwise stated.

The maximum allowable absolute and symmetrical deviations for control surfaces, slat and flap deflections are given in Table 7.

These values are applicable to the surfaces either at neutral positions or at their maximum deflections, unless otherwise stated.

Both absolute and symmetrical deviations have to be respected at the same time.

TABLE 7 : MAXIMUM ALLOWABLE ABSOLUTE AND SYMMETRY DEVIATIONS FOR CONTROL SURFACE, SLAT AND FLAP DEFLECTIONS

SURFACE	ABSOLUTE DEVIATION (DEG)	SYMMETRY DEVIATION (DEG)
AILERONS (2)	$\pm 0,5$	$\pm 1,0$
SPOILERS	$\pm 1,0$	$\pm 1,0$
HORIZONTAL STABILIZERS	$\pm 0,5$	$\pm 0,5$
ELEVATORS	$\pm 1,0$	$\pm 1,0$
RUDDER	$\pm 1,0$	N.A. (1)
TAKE-OFF SLATS	$\pm 0,5$	$\pm 1,0$
LANDING SLATS	$\pm 1,0$	$\pm 1,0$
RETRACTED FLAPS	$\pm 0,5$	$\pm 1,0$
TAKE-OFF FLAPS, MAIN ELEMENT, INBOARD AND OUTBOARD	$\pm 1,0$	$\pm 1,0$
TAKE-OFF FLAPS, AFT ELEMENT, INBOARD	$\pm 1,0$	$\pm 1,0$
LANDING FLAPS, MAIN ELEMENT, INBOARD AND OUTBOARD	$\pm 1,5$	$\pm 1,0$
LANDING FLAPS, AFT ELEMENT, INBOARD	$\pm 1,5$	$\pm 1,0$

(1) N.A.: Not Applicable.

(2) The maximum allowable absolute and symmetry deviations for the aileron at the neutral position are, respectively: $\pm 0,2^\circ$ and $\pm 0,4^\circ$.

NE 02-046

AERODYNAMIC SMOOTHNESS – REQUIREMENTS FOR ERJ 190 AND ERJ 195

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 17 of 22

5.6 COMPONENT ANGULAR TOLERANCES

The angular deviations of aircraft components may be divided into two types:

- Absolute Deviation: Angular difference between the actual and nominal position.
- Symmetry or Inter-Component Deviation: Angular difference between symmetrical stations (for example: port and starboard wing, pylons, horizontal tail) or between components of the same part (for example: front, center and rear fuselage).

The component angular tolerances are given in Table 8.

TABLE 8 : MAXIMUM ALLOWABLE COMPONENT ANGULAR DEVIATIONS

AIRCRAFT COMPONENT	MEASURED ANGLE	ABSOLUTE DEVIATION (minute)	SYMMETRY OR INTER-COMPONENT DEVIATION (minute)
WING	INCIDENCE AT ROOT	±10	±20
	ROOT-TIP TWIST	±10	±20
	DIHEDRAL	±20	±20
	SWEEP	±20	±20
WINGLET	INCIDENCE AT ROOT	±30	±30
	ROOT-TIP TWIST	±30	±30
	DIHEDRAL	±60	±60
	SWEEP	±60	±60
HORIZONTAL STABILIZER	INCIDENCE AT ROOT	±10	±20
	ROOT-TIP TWIST	±10	±20
	DIHEDRAL	±20	±20
	SWEEP	±20	±20
VERTICAL TAIL	LATERAL INCIDENCE AT ROOT	±10	N.A. (1)
	ROOT-TIP TWIST	±10	N.A. (1)
	VERTICAL ALIGNMENT	±15	N.A. (1)
	SWEEP	±20	N.A. (1)
PYLONS	INCIDENCE (ELEVATION)	±10	±10
	INCIDENCE (AZIMUTH)	±10	±10
	VERTICAL ALIGNMENT	±20	±20
FUSELAGE	ALIGNMENT OF SECTIONS WITH LONGITUDINAL AXIS	±10	±10
ENGINE	LONGITUDINAL ALIGNMENT	±10	±10
	LATERAL ALIGNMENT	±10	±10
BLADE ANTENNAS	NOMINAL INCIDENCE	±60	N.A. (1)

(1) N.A. = Not Applicable.

5.7 SPECIAL AERODYNAMIC REQUIREMENTS

Special aerodynamic smoothness requirements or exceptions to the values established on this Standard shall be specified on the production drawings, after submitting to analysis and approval by the Embraer Aerodynamic Group.

NE 02-046		AERODYNAMIC SMOOTHNESS – REQUIREMENTS FOR ERJ 190 AND ERJ 195	
REV: K	ISSUE DATE: 30 Jun 26	REVISION REPORT: RR_NE 02-046K	PAGE: 18 of 22

6. MAXIMUM ALLOWABLE DEFORMATIONS DUE TO IN-FLIGHT LOADS

The tolerances established in the previous items do not consider the aircraft components deformation that occur during normal flight. Based on wind tunnel tests and computation estimative, the aerodynamic deformation requirements are referred to as the nominal aircraft elements configuration and may be assumed as follows:

The sign convention “+” means an increase of the value of each of the items listed in the Table 9 and Table 10, due to in-flight loads. On the other hand, the sign convention “-” means the opposite.

TABLE 9 : MAXIMUM ALLOWABLE LINEAR DEFORMATIONS DUE TO IN-FLIGHT LOADS

ITEM	CONFIGURATION	MINIMUM (mm)	MAXIMUM (mm)	FLIGHT CONDITION
GAP BETWEEN SLATS AND FIXED LEADING EDGE	TAKEOFF POSITION	-2,0	+1,0	Vs for each slat position
GAP BETWEEN SLATS AND FIXED LEADING EDGE	LANDING POSITION	-2,0	+2,0	Vs for each slat position
OVERLAP BETWEEN SLATS AND FIXED LEADING EDGE	TAKEOFF POSITION	-2,0	+2,0	Vs for each slat position
OVERLAP BETWEEN SLATS AND FIXED LEADING EDGE	LANDING POSITION	-2,0	+2,0	Vs for each slat position
STEP BETWEEN SLAT TRAILING EDGE AND WING	RETRACTED POSITION	N.A. (1)	+0,5	Nz = 1g , Vc
GAP BETWEEN MAIN FLAP ELEMENT AND SPOILERS	ALL DEFLECTED POSITION	-2,0	+2,0	Vs for each flap position
GAP BETWEEN AFT AND MAIN FLAP ELEMENTS	ALL DEFLECTED POSITION	-2,0	+2,0	Vs for each flap position
OVERLAP BETWEEN MAIN FLAP ELEMENT AND SPOILERS	ALL DEFLECTED POSITION	-2,0	+3,0	Vs for each flap position
OVERLAP BETWEEN AFT AND MAIN FLAP ELEMENTS	ALL DEFLECTED POSITION	-2,0	+3,0	Vs for each flap position
SPOILER LEADING EDGE	RETRACTED FLAPS	N.A. (1)	+0,5	Nz = 1g , Vc
SPOILER LEADING EDGE	ALL DEFLECTED POSITION	N.A. (1)	+1,0	Vs for each flap position
SPOILER TRAILING EDGE	RETRACTED FLAPS	N.A. (1)	+2,0	Nz = 1g , Vc
SPOILER TRAILING EDGE	TAKEOFF FLAPS DEPLOYED	N.A. (1)	+2,0	Vs for each flap position
SPOILER TRAILING EDGE	LANDING FLAPS DEPLOYED	N.A. (1)	+5,0	Vs for each flap position
LOWER SHROUD TRAILING EDGE	RETRACTED FLAPS	N.A. (1)	+2,0	Nz = 1g , Vc
LOWER SHROUD TRAILING EDGE	TAKEOFF FLAPS DEPLOYED	N.A. (1)	+2,0	Vs for each flap position

The Table 9 continue.

NE 02-046

AERODYNAMIC SMOOTHNESS – REQUIREMENTS FOR ERJ 190 AND ERJ 195

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 19 of 22

The Table 9 continued.

TABLE 9 : MAXIMUM ALLOWABLE LINEAR DEFORMATIONS DUE TO IN-FLIGHT LOADS

ITEM	CONFIGURATION	MINIMUM (mm)	MAXIMUM (mm)	FLIGHT CONDITION
FLAP TRACK FAIRINGS SURFACE NORMAL DEVIATION RELATIVE TO THE SYSTEM LINE	RETRACTED POSITION	-3,0	+3,0	$N_z = 1g$, V_c
MAIN LANDING GEAR DOOR GAP	RETRACTED POSITION	-6,0	+6,0	$N_z = 1g$, V_c
MAIN LANDING GEAR DOOR STEP	RETRACTED POSITION	-3,0	+3,0	$N_z = 1g$, V_c
NOSE LANDING GEAR DOOR GAP	RETRACTED POSITION	-6,0	+6,0	$N_z = 1g$, V_c
NOSE LANDING GEAR DOOR STEP	RETRACTED POSITION	-3,0	+3,0	$N_z = 1g$, V_c

(1) N.A. = Not Applicable.

TABLE 10 : MAXIMUM ALLOWABLE ANGULAR DEFORMATIONS DUE TO IN-FLIGHT LOADS

ITEM	CONFIGURATION	MINIMUM (deg)	MAXIMUM (deg)	FLIGHT CONDITION
WING TWIST FROM ROOT TO BREAK	CRUISE	-0,5	+0,5	$N_z = 1g$, V_c
WING TWIST FROM BREAK TO TIP	CRUISE	-1,0	+0,5	$N_z = 1g$, V_c
SLAT DEFLECTION	ALL DEFLECTED POSITIONS	-0,5	+0,5	V_s for each slat position
MAIN FLAP ELEMENT	ALL DEFLECTED POSITIONS	-0,5	+1,0	V_s for each flap position
AFT FLAP ELEMENT	ALL DEFLECTED POSITIONS	-0,5	+1,0	V_s for each flap position
WINGLET TWIST	CRUISE	-0,5	+0,5	1) $N_z = 1g$, V_c 2) 1.2 V_s TO 3) climb
WINGLET DIHEDRAL	CRUISE	-2,0	+2,0	1) $N_z = 1g$, V_c 2) 1.2 V_s TO 3) climb
AILERON ABSOLUTE DEFLECTION	CRUISE	-0,5	+0,5	$N_z = 1g$, V_c
AILERON DEFLECTION SYMMETRY	CRUISE	-0,5	+0,5	$N_z = 1g$, V_c
HORIZONTAL TAIL TWIST	CRUISE	-0,5	+0,5	$N_z = 1g$, V_c
ELEVATOR ABSOLUTE DEFLECTION	CRUISE	-0,5	+0,5	$N_z = 1g$, V_c

The Table 10 continue.

NE 02-046

**AERODYNAMIC SMOOTHNESS –
REQUIREMENTS FOR ERJ 190 AND ERJ 195**

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 20 of 22

The Table 10 continued.

TABLE 10 : MAXIMUM ALLOWABLE ANGULAR DEFORMATIONS DUE TO IN-FLIGHT LOADS

ITEM	CONFIGURATION	MINIMUM (deg)	MAXIMUM (deg)	FLIGHT CONDITION
ELEVATOR DEFLECTION SYMMETRY	CRUISE	-0,5	+0,5	$N_z = 1g$, V_c
VERTICAL TAIL TWIST	CRUISE	-0,5	+0,5	$N_z = 1g$, V_c
RUDDER ABSOLUTE DEFLECTION	CRUISE	-0,5	+0,5	$N_z = 1g$, V_c
RUDDER TWIST	CRUISE	-0,5	+0,5	$N_z = 1g$, V_c

NOTES

- 1)** Angular deformations are considered as average values taken from five span wise stations representative of each movable panel.
- 2)** The aerodynamics requirements for gaps and steps of the passenger, service and cargo doors and all inspection doors and access panels during flight shall be according to the Table 3 and Table 4.
- 3)** Some aircraft components are required not to have buckling during flight. The Table 11 enumerates them and establishes the respective flight conditions.

TABLE 11 : FLIGHT CONDITIONS WHERE BUCKLING IS NOT ALLOWED

ITEM	FLIGHT CONDITION
WING	1) $N_z = 1,3g$, V_c 2) $N_z = 1,0g$, 1,23 V_s for each flap condition
SLAT	
FLAP	
WINGLET	1) $N_z = 1,3g$, V_c 2) 1,2 V_s TO
SPOILER	1) $N_z = 1,3g$, V_c 2) $N_z = 1,0g$, 1,23 V_s for each flap condition
LOWER SHROUD	
AILERON	1) $N_z = 1,3g$, V_c , neutral position 2) $N_z = 1,0g$, 1,23 V_s for each flap condition, full deflection
HORIZONTAL TAIL	
ELEVATOR	
VERTICAL TAIL	1) $N_z = 1,3g$, V_c 2) V_s for each flap condition
RUDDER	1) $N_z = 1,3g$, V_c , neutral position 2) $N_z = 1,0g$, 1,23 V_s for each flap condition, full deflection
FTF	

NE 02-046

**AERODYNAMIC SMOOTHNESS –
REQUIREMENTS FOR ERJ 190 AND ERJ 195**

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 21 of 22

7. QUALITY ASSURANCE PROVISIONS

7.1 The individual responsible for performing aerodynamic tolerance compliance shall be trained and shall guarantee the fulfillment of the requirements of this Standard.

7.2 Embraer Quality Assurance shall monitor the process of aerodynamic surface fabrication by an auditing process so as to guarantee the fulfillment of item 6.1 and promote preventive actions when necessary.

8. GENERAL NOTES

8.1 DRAWING CALLOUT

This Standard shall be called out on the skin assembly drawings for the fuselage, wing, horizontal and vertical tails and pylons fairing, as well as on the drawings related to the entire assemblies, through the drawing note **below**:

NI 1234 - Aerodynamic Smoothness, according to NE 02-046.

9. BIBLIOGRAPHY

This Standard has been based on the following documents:

9.1 Appendix IIC: Nacelle Aerodynamic Smoothness Requirements of Section 2. Power-plant Specification of Document CF34-8E Power-plant System for Embraer ERJ 170/ERJ 190 - Exhibit A1-1 Contractors Technical Proposal.

9.2 NE 02-041: Aerodynamic Smoothness - Requirements for EMB-145.

9.3 NE 02-044: Aerodynamic Smoothness - Requirements for ERJ 170.

9.4 190-GEA-AA-2001-04 : External Surface Aerodynamic Smoothness and Angular Tolerance Requirements for the ERJ 190 and ERJ 195 Aircraft.

9.5 314-EAD-VA97-01 : External Surface Aerodynamic Smoothness and Angular Tolerance Requirements for the EMB-314 Aircraft.

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NE 02-046

**AERODYNAMIC SMOOTHNESS –
REQUIREMENTS FOR ERJ 190 AND ERJ 195**

REV: K

ISSUE DATE: 30 Jun 26

REVISION REPORT: RR_NE 02-046K

PAGE: 22 of 22